

R Markdown + GitHub Workshop

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1 Before you start (one-time setup)

Please run the following in your R console **before starting the tasks**:

```
install.packages(c(
  "rmarkdown",
  "knitr",
  "palmerpenguins",
  "dplyr",
  "ggplot2"
))
```

If you are knitting to PDF, also run:

```
install.packages("tinytex")
tinytex::install_tinytex()
```

2 Starting with R Markdown: workshop tasks

2.1 Task 1: Open the file

Open the file `RMarkdown_Intro.Rmd`.

Change the title of the Markdown document from:

My First Markdown Document

To:

RMarkdown Introduction

Click the **Knit** button to see the compiled version of your sample code.

Congratulations! You've just knitted your first R Markdown document!

2.2 Task 2: Basic text editing

1. Change the author to your own name
 2. Rewrite the first sentence to:
`This is my first R Markdown document.`
 3. Recompile the document
-

2.3 Task 3: Adding a link

You can turn a word into a link like this:

```
[RStudio] (https://www.rstudio.com)
```

2.3.1 Your task

Edit the text below so that **school's GitHub page** becomes a clickable link to:
<https://github.com/SoSS-UoM>

2.3.2 Text to edit

Material from today's workshop can be found on the school's GitHub page.

2.3.3 Expected result

```
Material from today's workshop can be found on the [school's GitHub page] (https://github.com,
```

2.4 Task 4: Text formatting

In this task, you will practise basic Markdown formatting to improve the appearance and readability of your document. You will apply *italic*, **bold**, and `monospaced` text, and organise information using a list.

- *italic* → **text**
- **bold** → ****text****
- `monospaced` → ``text``

2.4.1 Your task

Take the original text below and:

- Convert it into a **bullet point list**
- Apply **bold**, *italic*, and `monospaced` formatting where appropriate (feel free to experiment!)

2.4.2 Original text

When analysing data, a starting point is to examine the characteristics of each individual variable in the data set. The way to proceed depends upon the type of variable being examined. The variables can be one of two broad types:

Attribute variable: has its outcomes described in terms of its characteristics or attributes;
Measured variable: has the resulting outcome expressed in numerical terms.



Tip

Note: Add a blank line before starting your list, otherwise it may not render correctly.

2.5 Task 5: Embedding R code

In this task, you will explore how R code is included and executed in your document.

In the **Including R Code** section, you will see a code chunk where the option `eval = FALSE` is currently set. Code chunks in R Markdown begin with ````${code}```` and end with `````, and any code written between them will be treated as R code when the document is knitted.

2.5.1 Your task

1. Change `eval = FALSE` to `eval = TRUE`.
2. Knit the document.

What happens when the code is executed?

Now look at the code chunk again.

3. Change `echo = TRUE` to `echo = FALSE`.
4. Knit the document again.

What is different this time?



Tip

Think about:

- When is the code run?
- When is the code shown?
- When do you only see the output?

2.6 Task 6: Creating a plot and controlling its output

In this task, you will make a plot appear in your document and explore how code visibility works.

In the **Including Plots** section, the code chunk is currently set so that:

- the code is hidden: `echo = FALSE`
- the code does not run: `eval = FALSE`

2.6.1 Your task

1. Change `eval = FALSE` to `eval = TRUE`.
2. Knit the document.

What appears in your document?

3. Now change `echo = FALSE` to `echo = TRUE`.
4. Knit the document again.

What is different now?



Tip

Think about:

- What controls whether a plot appears?
- What controls whether the code is visible?
- When might you want to show the plot but hide the code?

2.7 Task 7: Adding mathematical equations

In this task, you will learn how to include mathematical equations in your document using LaTeX.

For more information about LaTeX, visit:

<https://www.latex-project.org/>

A helpful reference for common LaTeX maths symbols and commands can be found here:

<https://en.wikibooks.org/wiki/LaTeX/Mathematics>

In the **Including Mathematical Equations** section, you will see the equation:

$$y = a + bx$$

2.7.1 Your task

1. Find where the equation is written in your `.Rmd` file.
2. Make sure it is displayed on its own line using double dollar signs `$$` `$$`.
3. Knit the document.

How does the equation appear in the output?

Now try displaying the equation **inline** within a sentence.

4. Add the equation into a sentence using single dollar signs `$` `$`, for example:

The model can be written as `$y = a + bx$`.

5. Knit the document again.

What is the difference between inline and displayed equations?

 Tip

Think about:

- When would you use an inline equation?
 - When would you display an equation on its own line?
 - Try exploring symbols such as subscripts $_$, superscripts $^$, and commands like `\\alpha`, `\\beta`.
-

2.8 Git practice

Try to get into the habit of saving your work using [Git](#) as you go.

Let's explore how to do this using [Git Desktop](#).

```
git status
git add RMarkdown_Intro.Rmd
git commit -m "Complete Task X"
```

At the end of the workshop, push your changes to GitHub:

```
git push
```